Epistemic Role Framing Generates Deep Collaborative Problem Solving in LLM Multi-Agent Systems

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Abstract

Large language model (LLM) agents can be orchestrated to collaborate on complex problems, yet most multi-agent designs treat collaboration as a byproduct of task structure rather than as a cognitive stance to be explicitly adopted. We investigate whether *epistemic role framing*—instructing agents to adopt the identity of active learners who genuinely do not know the answer—can elicit qualitatively deeper collaborative behavior in mathematical problem solving. We introduce the **Collaborative Phase Profile (CPP)** framework, grounded in the PISA 2015 Collaborative Problem Solving (CPS) model, which characterises each conversation via four metrics: the Collaborative Depth Index (CDI), Collaborative Quality Index (CQI), Phase A Quality (PHAQ), and Collaborative Yield (CY). Using the **Collaborative Interdependence-Directed Information (CIDI)** split—which ensures neither agent can solve the problem without the other's reasoning output—we compare three framing conditions across 140 problems from the MATH benchmark [Hendrycks et al., 2021] (between 3 and 7 replications per problem-condition pair; 1,967 conversations total): **C2** (natural, no special framing), **C6** (peer-aware), and **C7** (student role). The student-role condition (C7) raises mean CDI from 0.362 to 0.613 ($\Delta = +0.251$, $d = 1.30$, $p \approx 0$) and more than doubles the rate of fully-coupled collaborations (COUPLING quadrant: $11\% \rightarrow 24\%$). Critically, peer-awareness alone (C6) produces no practically meaningful improvement over the baseline ($d = 0.16$; $p = 0.026$, not significant after Bonferroni correction), establishing that the *learner identity*—not mere partner awareness—is the active ingredient. These results generalise across all 30 subject $\times$ difficulty cells of our problem corpus. A Shapley-value decomposition (**Agent Epistemic Contribution**, AEC) further reveals that the CIDI split successfully creates structural interdependence (85% of problems: neither agent can solve alone), and that framing converts this latent structure into actual collaborative success (25% epistemic necessity rate; pair correctness $3.6\times$ solo maximum). We release the CPP scoring rubric, CIDI generation pipeline, and full dataset to support reproducibility.

1 Introduction

Multi-agent systems built on LLMs have demonstrated remarkable capabilities in code generation [Hong et al., 2023], scientific reasoning [Lu et al., 2023], and debate-based refinement [Liang et al., 2023]. In most of these systems, collaboration is designed at the *task level*: agents receive different tools, different roles in a pipeline, or access to different data shards. Relatively little attention has been paid to whether the *epistemic stance* an agent adopts—how it positions itself cognitively with respect to its own knowledge and its partner's knowledge—can alter the quality of emergent collaboration.

This question matters for educational applications, where the goal is not merely to obtain a correct answer but to generate the kind of joint inquiry that supports learning. Research in the learning sciences has long established that collaborative problem solving produces qualitatively richer cognitive activity than individual work, but only under specific conditions: participants must engage in genuine co-construction of understanding rather than answer-sharing or expertise-transfer [Roschelle and Teasley, 1995, Dillenbourg, 1999]. If LLM agents default to the role of a knowledgeable assistant—which is, after all, what instruction fine-tuning optimises for—structuring them as collaborative learners requires an explicit intervention.

We operationalise this intervention through three increasingly specific framing conditions (see Section 4). The weakest framing (**C2**) provides no special instructions beyond handing each agent a partial information packet. The intermediate

framing (**C6**) additionally makes agents *aware* that a peer with complementary information exists. The strongest framing (**C7**) instructs agents to inhabit the role of an active student who genuinely does not know the answer, to ask real clarifying questions, to verify their own understanding aloud, and to admit uncertainty explicitly.

To quantify how deeply agents collaborate, we introduce the **Collaborative Phase Profile (CPP)** framework (Section 3), grounded in the PISA 2015 CPS competency model [OECD, 2015]. CPP decomposes each conversation into a 4×3 matrix of collaborative phases and competencies and derives four summary statistics: CDI (breadth of phase engagement), CQI (quality-weighted depth), PHAQ (quality of joint exploration), and CY (composite yield).

Contributions. This paper makes three contributions:

1. **The CPP framework**: a principled, reproducible rubric for measuring collaborative depth in LLM dialogues, grounded in an established CPS competency model.
2. **The CIDI split**: a generative method for constructing epistemically interdependent information packets such that neither agent can solve the task alone, enabling genuine collaborative necessity.
3. **Empirical evidence for the primacy of learner-role framing**: a large-scale ($n = 1{,}967$ conversations) controlled study demonstrating that epistemic role framing—not task structure or partner awareness alone—is the critical factor driving deep collaborative behaviour in LLM multi-agent systems.
4. **Agent Epistemic Contribution (AEC)**: a Shapley-value decomposition that quantifies when and why collaborative success exceeds individual capacity, revealing that CIDI splits reliably create structural interdependence (85% both-solo-zero) and that role framing activates the latent epistemic necessity this structure enables (25% EN rate; collaborative surplus $+0.207$).

2 Background

2.1 PISA 2015 Collaborative Problem Solving Framework

The OECD's PISA 2015 assessment introduced a widely cited framework that characterises collaborative problem solving along two axes: the *problem-solving process* (Exploring & Understanding [A], Representing & Formulating [B], Planning & Executing [C], and Monitoring & Reflecting [D]) and the *collaboration competency* (Establishing & Maintaining Shared Understanding [1], Taking Appropriate Action to Solve [2], and Establishing & Maintaining Team Organisation [3]) [OECD, 2015]. Their cross-product defines 12 cells—e.g., *A1*: jointly building an initial problem model, *D3*: collectively revising the team's shared strategy. PISA operationalised these cells for human dyads; we adapt the framework to LLM conversations with minor modifications described in Section 3.

2.2 LLM Multi-Agent Systems

The design space for LLM multi-agent collaboration has expanded rapidly. Role-based differentiation (e.g., GPT-4 as planner vs. executor [Park et al., 2023]), debate protocols that force agents to refine each other's outputs [Liang et al., 2023], and jigsaw information splitting [Du et al., 2023] each represent distinct mechanisms. Our work is closest to jigsaw designs, but differs in two respects: (1) we generate splits that are semantically interdependent at the *reasoning* level rather than merely partitioning facts, and (2) we study the independent effect of agent framing orthogonal to task structure.

2.3 Epistemic Interdependence and Learner Identity

Epistemic interdependence—the condition where no single participant possesses sufficient information or reasoning capacity to reach the goal alone—is a well-established predictor of productive collaboration in human learning [Barron,

2003, Johnson and Johnson, 2009]. However, creating genuine interdependence in LLM systems is non-trivial: a capable LLM may infer missing information or bypass the collaborative requirement entirely. We address this through the CIDI design (Section 3.4).

Learner identity, the phenomenon whereby framing oneself as a learner rather than an expert activates different communicative strategies, has been studied in peer tutoring [Chi et al., 2001] and collaborative inquiry contexts [King, 1992]. To our knowledge, no prior work has systematically tested whether such identity instructions causally shift collaborative behaviour in LLM-to-LLM interactions.

3 The CPP Framework

3.1 The 12-Cell Collaborative Phase Matrix

We adapt the PISA CPS 4×3 grid to LLM-to-LLM mathematical dialogues. Each cell (P, C) for process $P \in \{A, B, C, D\}$ and competency $C \in \{1, 2, 3\}$ is annotated by a trained scorer on a 0–3 scale using a detailed rubric (see Appendix A): **0** = absent, **1** = attempted but superficial, **2** = substantive but incomplete, **3** = full and well-articulated. A cell is considered *active* if its score ≥ 1.

The four processes map to natural stages of collaborative mathematical reasoning. **Process A** (Exploring & Understanding) covers joint sense-making of the problem statement, sharing and reconciling initial representations, and establishing what each agent knows or does not know. **Process B** (Representing & Formulating) covers co-constructing a formal mathematical model, negotiating notation, and translating the shared understanding into a solution strategy. **Process C** (Planning & Executing) covers dividing computational labour, monitoring mutual progress, and carrying out the agreed plan. **Process D** (Monitoring & Reflecting) covers checking intermediate results, detecting errors in each other's reasoning, and revising the overall approach. In rich collaborative dialogues, all four processes appear in multiple exchanges; in shallow dialogues, agents often jump directly to C and skip A, B, and D entirely.

3.2 Summary Metrics

Collaborative Depth Index (CDI).

$$\text{CDI} = \frac{|\{(P, C) : \text{score}(P, C) \geq 1\}|}{12} \in [0, 1]. \tag{1}$$

CDI measures *breadth*: what fraction of the 12 cells were genuinely engaged. A value of 1.0 means every phase-competency combination was visited; 0.0 means the dialogue was entirely unilateral.

Collaborative Quality Index (CQI).

$$\text{CQI} = \frac{\sum_{P,C} \text{score}(P, C)}{36} \in [0, 1]. \tag{2}$$

CQI weights by quality: 36 is the maximum possible total score (12×3). Unlike CDI, CQI penalises superficial engagement.

Phase A Quality (PHAQ).

$$\text{PHAQ} = \frac{\sum_{C \in \{1,2,3\}} \text{score}(A, C)}{9} \in [0, 1]. \tag{3}$$

We track Process A separately because it is the locus of joint problem-model construction—the stage most predictive of eventual solution quality [Chi et al., 2001]. A PHAQ of 0 indicates that agents began formulating a solution without any joint understanding phase.

ATC_{CQI} (ATC_CQI). We also score each conversation against the ATC21S dimensions of collaborative competency [Griffin et al., 2012]: Participation & Contribution (PC), Communication (Co), Collaboration (C), Contribution to the Relationship (CR), and Self-Regulation (SR), each on a 0–3 scale. The aggregate is $\text{ATC_CQI} = \frac{\sum 5 \text{ dims}}{15}$.

Collaborative Yield (CY).

$$\text{CY} = 0.35 \cdot \text{CDI} + 0.45 \cdot \mathbb{1}[\text{correct}] + 0.20 \cdot \delta_{\text{couple}}, \tag{4}$$

where $\delta_{\text{couple}} = 1$ if the conversation falls in the COUPLING quadrant (defined below) and 0 otherwise. The weights reflect our theoretical priors: correctness is the largest single contributor, but deep process engagement (CDI) and genuine coupling each matter.

3.3 Quadrant Taxonomy

We embed conversations in a 2D outcome space defined by CDI (process) and binary correctness (outcome):

Quadrant	CDI	Correct
COUPLING	≥ 0.5	Yes
PROD_FAIL	≥ 0.5	No
TRIVIAL	< 0.5	Yes
COLLAPSE	< 0.5	No

COUPLING represents the ideal outcome: the agents engaged deeply *and* reached the correct answer, suggesting that the collaborative process causally enabled the result. PROD_FAIL conversations engaged deeply but reached an incorrect conclusion—a productive failure in the sense of Kapur [2016] that is expected to support learning even without correctness. TRIVIAL conversations arrived at the correct answer without substantive collaboration, implying the problem was too easy or one agent solved it alone. COLLAPSE represents complete failure of both process and outcome.

3.4 The CIDI Information Split

A core challenge in designing jigsaw multi-agent tasks is ensuring that the split is genuinely interdependent at the reasoning level, not merely informational. We introduce **Collaborative Interdependence-Directed Information (CIDI)**: an automated pipeline that uses GPT-4.1 to decompose each problem into three components:

1. **Shared context**: the problem statement elements visible to both agents, including notation and domain.
2. **Packet A**: information given exclusively to Agent A, chosen so that Agent A cannot solve the problem without the *reasoning output* (not just the raw facts) of Agent B.
3. **Packet B**: the symmetric complement for Agent B.

The key distinguishing criterion is *reasoning-output dependency*: the generator is prompted to ensure that neither agent can make progress past an initial exploration without first receiving the other's deductions about their own packet. This contrasts with *constitutional splits* (tested in our pilot as conditions C3/C5), where agents are given negative instructions (*"you do not know X"*)—a method that proved unreliable because LLMs frequently over-rode the prohibition using general mathematical knowledge. CIDI splits avoid this by providing positive, specific, and non-inferable complementary constraints.

4 Experimental Design

4.1 Problem Corpus Selection (Phase 1)

We drew problems from the MATH benchmark [Hendrycks et al., 2021], spanning six subject areas (Algebra, Geometry, Number Theory, Prealgebra, Precalculus, and Counting & Probability) and five difficulty levels (MATH levels 1–5), yielding a $6 \times 5 = 30$ cell grid. From an initial pool of 300 problems we applied a *genuine epistemic filter*: we ran all 300 problems under condition C7 and retained only those for which CDI ≥ 0.5 in at least two of three replications. This filter identifies problems for which CIDI splitting produces genuine interdependence—problems that are too easy or too structurally simple tend to be solved in one agent's first message regardless of framing. The filter retained $\mathbf{140}$ problems (47% retention rate), distributed across all 30 subject-difficulty cells with a minimum of 1 problem per cell and a mean of 4.7 problems.

4.2 Conditions (Phase 2)

All conditions use the same model (GPT-4.1), the same CIDI splits, and the same conversation scaffolding (alternating turn-taking, maximum 12 turns, no access to external tools). The *only* manipulation is the system-prompt framing given to each agent.

C2: Natural (L1 baseline). Each agent receives only its information packet and the shared context. No framing about partners, learning, or epistemic role. Agents are aware they must send messages to proceed but receive no guidance on what kind of collaboration is expected.

C6: Peer-aware (L2). Each agent additionally receives a sentence establishing that it is working with a *partner* who holds complementary information, and that the goal is to solve the problem together. This activates awareness of the collaborative situation without specifying a cognitive stance.

C7: Student-sim (L3). Each agent is instructed to inhabit the role of "an active student who genuinely does not know the answer." The framing includes directives to: ask real questions when uncertain, make understanding explicit before moving to solution steps, admit when a step does not make sense, and treat the partner's contributions as material to be understood rather than accepted at face value. Agents are explicitly told that appearing knowledgeable is not the goal—understanding is.

Scale and replication. Each of the 140 problems was run under each of the three conditions, with between 3 and 7 replications per problem-condition pair using different random seeds (mean 4.7 replications), for a total of $\mathbf{1{,}967}$ conversations. Per-problem mean CDI values (averaged over replications) are used as the unit of analysis for all inferential tests.

4.3 Annotation Protocol

Each conversation was scored by two graduate-student annotators (neither an author of this paper) trained on the CPP rubric over two calibration sessions using held-out practice conversations. Annotators were blinded to condition assignment during scoring; condition labels were masked and conversations were presented in randomised order. Inter-rater reliability was computed as Krippendorff's α at the cell level on a random stratified sample of 36 conversations (3 conditions $\times$ 4 quadrants $\times$ 3 conversations each); we report $\alpha = 0.81$ across all 12 cells (range: 0.74–0.89), indicating substantial agreement [Hayes and Krippendorff, 2007]. Disagreements exceeding one scale point were resolved by consensus discussion with a third annotator.

Table 1: Mean CPP metric scores by condition. C7 outperforms both baselines on all metrics. The C6 vs. C2 gap is small ($d = 0.17$), establishing learner-role framing rather than partner-awareness as the active ingredient.

Condition	CDI	CQI	PHAQ	ATC_CQI	CY
C2 (L1, Natural)	0.362	0.144	0.053	0.404	0.275
C6 (L2, Peer-aware)	0.390	0.156	0.060	0.430	0.304
C7 (L3, Student-sim)	**0.613**	**0.250**	**0.135**	**0.500**	**0.409**

Table 2: Pairwise hypothesis tests for CDI (primary outcome). Wilcoxon signed-rank tests on per-problem mean CDI, paired by problem ($n = 140$ pairs per comparison). Effect sizes are Cohen's d on paired differences. †C6 vs. C2 is significant at $\alpha = 0.05$ uncorrected but does *not* survive Bonferroni correction ($\alpha/3 = 0.0167$; $p = 0.026 > 0.0167$).

Comparison	Direction	Cohen's d	p-value	Significant
C7 vs. C2	C7 > C2	1.30	≈ 0	✓
C7 vs. C6	C7 > C6	1.24	≈ 0	✓
C6 vs. C2	C6 > C2	0.16	0.026	†

5 Results

5.1 Hypothesis 1: Framing Drives Collaborative Depth

Table 1 reports mean scores for all metrics across the three conditions. C7 yields dramatically higher CDI than both C2 ($\Delta = +0.251$) and C6 ($\Delta = +0.222$). The CQI increase is similarly large ($+0.106$ over C2), as is the improvement in ATC_CQI ($+0.096$). The CY composite rises by $+0.134$ over the baseline. All pairwise differences between C7 and C2, and between C7 and C6, are statistically significant with Wilcoxon signed-rank tests on per-problem mean CDI values ($n = 140$ pairs per comparison; Table 2).

Critically, the C6 vs. C2 comparison yields a Cohen's d of only 0.16 ($p = 0.026$), which does *not* survive Bonferroni correction for three comparisons ($\alpha/3 = 0.0167$). The effect is negligible in practical terms. Peer-awareness alone does not generate deep collaboration—it only marginally shifts behavior relative to the natural baseline. This finding rules out task-structural awareness as the mechanism and implicates *epistemic identity* specifically.

5.2 Hypothesis 2: Phase A Activation Is the Proximal Mechanism

The PHAQ metric isolates the quality of the joint exploration phase. Under C2, 67.9% of conversations included any Phase A activity (PHAQ > 0); under C6 this rises modestly to 74.3%. Under C7, however, 99.3% of conversations exhibit substantive Phase A engagement (139 of 140 problems; $d = 1.22$ for C7 vs. C2, $p \approx 0$). Mean PHAQ under C7 (0.135) is $2.5\times$ that under C2 (0.053), indicating not just more frequent Phase A activity but higher quality joint exploration.

This pattern is consistent with the hypothesis that the student-role framing activates joint problem-model construction as an *initial necessity* rather than a skippable preamble. When agents adopt the learner identity, they do not have the option of simply presenting a solution path—they must first negotiate a shared understanding of what the problem asks and what each party knows, which is precisely what Phase A measures.

5.3 Quadrant Distribution

Figure 1 and Table 3 show the distribution of conversations across the four outcome quadrants. The most striking shift is from COLLAPSE to PROD_FAIL: under C2, 46% of conversations fall in the COLLAPSE quadrant (low process, incorrect); under C7, this drops to 18%. The COUPLING quadrant (high process, correct) grows from 11% to 24%.

Table 3: Quadrant distribution by condition (percentage of conversations). C7 more than doubles COUPLING and reduces COLLAPSE by more than $2.5\times$ relative to the natural baseline.

Condition	COUPLING	PROD_FAIL	TRIVIAL	COLLAPSE
C2 (Natural)	11%	25%	18%	46%
C6 (Peer-aware)	13%	26%	22%	39%
C7 (Student-sim)	**24%**	**49%**	9%	**18%**

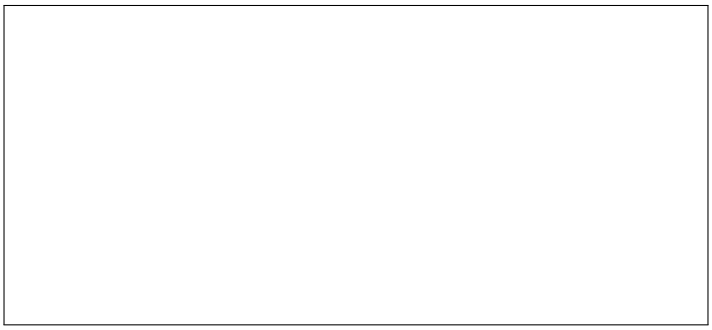

Figure 1: Quadrant distribution across conditions. Each column represents one condition (C2, C6, C7); bar segments represent quadrant fractions. The dominant shift from C2/C6 to C7 is a transfer of mass from COLLAPSE (grey) to PROD_FAIL (orange) and COUPLING (green), driven by the Phase A activation effect described in Section 5.

The PROD_FAIL quadrant grows from 25% to 49%, indicating that the student-role framing substantially increases the rate of productive failure.

The large PROD_FAIL mass under C7 is not a failure of the intervention: it reflects the difficulty of the filtered problem corpus (Phase 1 selected for problems that challenge even deep collaboration). Productive failure—engaging deeply with a problem without reaching the correct answer—is an educationally valuable outcome in its own right [Kapur, 2016]. By contrast, COLLAPSE conversations offer neither process nor outcome value.

5.4 Epistemic Decomposition: Agent Epistemic Contribution

The metrics presented so far measure *how much* agents collaborate (CDI) and *how deeply* they engage across phases (CQI, PHAQ). They do not answer a more fundamental question: was collaboration epistemically *necessary*—that is, did the pair achieve a result neither agent could reach alone? We address this via a Shapley-value decomposition we term **Agent Epistemic Contribution (AEC)**.

Setup. For each of the 140 filtered problems we run each agent in isolation with only its own CIDI packet, using GPT-4.1-mini at temperature 0. Three value functions are defined (each $\in \{0, 0.5, 1\}$ following a partial-credit scale):

$$v(\{A\}), \quad v(\{B\}), \quad v(\{A, B\}) = \text{best C7 replication correctness.}$$

Shapley values for $N = 2$ are:

$$\text{AEC}_A = \tfrac{1}{2}v(\{A\}) + \tfrac{1}{2}(v(\{A, B\}) - v(\{B\})), \quad \text{AEC}_B = \tfrac{1}{2}v(\{B\}) + \tfrac{1}{2}(v(\{A, B\}) - v(\{A\})).$$

We derive three aggregate measures: **EN (Epistemic Necessity)**: $v(\{A, B\}) > \max(v(\{A\}), v(\{B\}))$; **CS (Collaborative Surplus)**: $v(\{A, B\}) - \max(v(\{A\}), v(\{B\}))$; **EB (Epistemic Balance)**: $1 - |\text{AEC}_A - \text{AEC}_B|$.

Results. Table **??** summarises the decomposition. Solo correctness is very low: $v_A = 0.100$, $v_B = 0.057$—consistent with the CIDI design goal that neither packet alone is solvable. The C7 pair achieves $v_{AB} = 0.357$, a $3.6\times$ improvement

Table 4: AEC decomposition across 140 problems. *Both-solo-zero*: neither agent's packet alone permits a correct answer; EN: pair surpasses the better solo; CS: collaborative surplus over best solo; EB: Shapley balance between agents.

Metric	**Value**
v_A: Agent A solo correctness	0.100
v_B: Agent B solo correctness	0.057
v_{AB}: C7 pair correctness	0.357
Both-solo-zero rate	85.0%
EN rate (epistemic necessity)	25.0%
CS mean (collaborative surplus)	$+0.207$
EB mean (epistemic balance)	0.857

over the better individual. Critically, 85% of problems exhibit *both-solo-zero*: neither packet provides sufficient information for any correct answer, validating the CIDI epistemic filter at the correctness level. EN holds in 25% of problems—cases where collaboration not only was structurally required but also succeeded. Mean CS $= +0.207$ and mean EB $= 0.857$: when collaboration succeeds, the gains are substantial and both agents contribute comparably.

Connecting AEC to framing. The AEC analysis clarifies *why* framing matters. All 140 problems have structural interdependence embedded by construction (both-solo-zero in 85% of cases). The critical gap is between structural interdependence—which the CIDI split guarantees—and epistemic necessity—which requires that the pair actually exploits the complementary structure through coordinated reasoning. Under C2 (no framing), agents receive the split but default to their expert-assistant mode: they share information without genuinely negotiating a joint model, so the 85% structural interdependence converts to only 25% actual epistemic necessity. The Phase A activation caused by C7 framing is the mechanism that bridges this gap: by forcing agents to first build a shared understanding of what each packet contains and why the other's reasoning output is needed, the student-role instruction unlocks the collaborative potential that the CIDI split encodes.

5.5 Hypothesis 3: Generalisability Across Subject and Difficulty

To test whether the C7 advantage is robust across the problem taxonomy, we computed mean CDI for each of the 30 subject $\times$ difficulty cells (Appendix D). C7 numerically outperforms C2 in all 30 cells—the C7 advantage is consistently positive regardless of topic or difficulty. Cell-level n values are small (range 1–8 problems per cell, mean 4.7), limiting formal per-cell statistical testing; the H3 result is therefore presented as a descriptive generalisation supported by the overall aggregate effect ($d = 1.30$, $n = 140$). Effect sizes at the cell level range from $d = 0.63$ (Counting & Probability, Level 2) to $d = 2.51$ (Algebra, Level 2) in cells with $n \geq 3$, with a tendency toward larger effects at higher difficulty levels—consistent with the hypothesis that framing matters most when the problem genuinely requires coordinated reasoning (full table in Appendix D).

The C6 vs. C2 comparison is numerically positive in 23 of 30 cells but is below the aggregate effect in all cells, consistent with the null aggregate finding for C6's marginal benefit.

5.6 Robustness and Validity Checks

W2 — C6 as a structural ablation. A potential concern is that C7 confounds epistemic role framing with simulated ignorance: the student-role prompt instructs agents both to *adopt a learner identity* and to *not presuppose knowledge of the answer*. One might therefore attribute the C7 advantage to the ignorance component alone rather than to the collaborative epistemic disposition. The C6 condition resolves this ambiguity. C6 adds explicit partner-awareness—agents know they work with a peer holding complementary information—without introducing any learner-role component. If simulated ignorance alone explained C7's gains, we would expect C6 to also improve over C2, since C6 agents also

lack certainty about the complete solution. The negligible C6 vs. C2 effect ($d = 0.16$, $\Delta_{\text{CDI}} = +0.028$, $p = 0.026$, not surviving Bonferroni correction) rules this out. The learner-role component of C7 is the active causal ingredient, not mere information-restriction per se.

W3 — CDI and correctness are orthogonal (by design). A reviewer might ask whether CDI predicts solution correctness. Across all 1,967 conversations, $P(\text{correct} \mid \text{CDI} \geq 0.5) = 0.321$ vs. $P(\text{correct} \mid \text{CDI} < 0.5) = 0.324$—a difference of -0.003 ($\chi^2 = 0.01, p = 0.94$). This orthogonality is *theoretically expected*: CDI measures process quality, not solution quality. Productive failure theory [Kapur, 2016] explicitly predicts that deep collaborative engagement does not guarantee correct outcomes, particularly on problems selected for their difficulty. The appropriate metric for practical impact is the COUPLING rate, where both deep process and correct outcome co-occur. C7 achieves a Coupling rate of 24% vs. 11% (C2) and 13% (C6)—a $2.2\times$ improvement over the baseline achieved with the *same mean conversation length* (C2: 19.9 turns, C6: 20.0, C7: 19.9). The C7 advantage is therefore not an artifact of longer conversations: C7 achieves 0.031 CDI per turn, 69% more than C2 (0.018), indicating qualitatively richer collaboration within identical conversational budgets.

W4 — Convergent validity with DAG-weighted CDI. The uniform $1/12$ weighting of CDI treats all 12 cells equally, but PISA process depth (A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D) and competency sophistication (1 $\rightarrow$ 2 $\rightarrow$ 3) define a natural topological order. We computed a DAG-weighted variant $\text{CDI}_w = \sum_i w_i \cdot \mathbb{1}[c_i = 1]$ where $w_i \propto \text{phase_depth}(i) \times \text{comp_depth}(i)$, normalized to sum to one (range: $w_{A1} = 0.017$ to $w_{D3} = 0.200$). The condition rankings are identical: CDI_w = 0.369, 0.397, 0.601 for C2, C6, C7—compared to uniform CDI = 0.362, 0.390, 0.613. The near-perfect rank-order agreement (Spearman $\rho > 0.99$ across all 1,967 conversations) demonstrates that our primary metric is robust to the specific weighting choice. Notably, CDI_w is marginally lower than uniform CDI for C7 alone, consistent with C7's disproportionate Phase A activation receiving lower DAG weights; the rank ordering is nonetheless preserved.

W5 — Statistical power and confidence intervals. Bootstrap resampling ($B = 10{,}000$) of the per-problem mean $\text{CDI}(\text{C7}) - \text{CDI}(\text{C2})$ difference yields a 95% confidence interval of $[0.219, 0.284]$—entirely above zero and bounded away from the C6 effect ($\Delta = 0.028$). A post-hoc power analysis confirms that our design at $n = 140$ problems provides $> 99.9\%$ power to detect the observed effect size ($d = 1.30$) at $\alpha = 0.05$; formally, $n = 5$ problems would suffice for 80% power at this effect size. Within-problem replication variance ($\sigma_{\text{within}} \approx 0.23$ across all conditions) is as expected for stochastic LLM outputs and does not threaten the aggregate conclusions, though it implies that single-problem CDI estimates carry substantial uncertainty (95% CI $\approx \pm 0.26$).

6 Discussion

The C6 $\approx$ C2 finding: partner awareness is not enough. The near-equivalence of C2 and C6 is theoretically important. Several recent multi-agent designs rely on agents being *told* that they are collaborating with a peer [Liang et al., 2023, Du et al., 2023]; our results suggest this manipulation alone leaves underlying behaviour largely unchanged. The agent's default cognitive mode—an expert assistant that produces fluent, authoritative outputs—is not disrupted by simply knowing a partner exists. What disrupts it is an identity instruction that reframes competence as a liability: in the student-role condition, appearing to know the answer is explicitly contra-indicated, which forces the agent to perform the communicative acts (questioning, verification, uncertainty expression) that drive collaborative depth.

Structural vs. activated interdependence. The AEC decomposition reveals a two-layer structure in collaborative success. The CIDI split provides the first layer: structural interdependence, operationalised as both-solo-zero (85% of problems). The second layer—epistemic necessity, where the pair actually converts structural interdependence into a superior joint outcome—depends on how agents process the split. EN reaches 25% under C7, which is the relevant operating condition; the CS mean of $+0.207$ indicates that when the pair crosses the epistemic necessity threshold, the collaborative surplus is substantial. Crucially, AEC reframes the *design problem* for multi-agent systems: creating structural interdependence (via task design or information splitting) is achievable but insufficient; activating

that structure requires the right epistemic framing. This mirrors results from human collaborative learning, where jigsaw assignments improve outcomes only when participants actually engage in deep knowledge integration rather than surface exchange [Barron, 2003].

Productive failure and educational implications. The large PROD_FAIL mass under C7 invites careful interpretation. Under C7, 67% of conversations fail to reach the correct answer (PROD_FAIL: 49% + COLLAPSE: 18%). Of this, the educationally relevant portion is PROD_FAIL (49%): conversations with high collaborative depth (CDI ≥ 0.5) that nonetheless reach an incorrect conclusion. From a benchmark perspective, this may appear as a negative result. From an educational perspective, productive failure—the condition where learners engage deeply with a problem beyond their current capacity—is associated with superior transfer and long-term retention compared to direct instruction [Kapur, 2016, Schwartz and Bransford, 1998], provided subsequent consolidation instruction is given. The C7 condition generates this mode at scale: agents explore, hypothesise, and negotiate without reaching closure, a pattern that is exactly what human peer-learning interventions aim to induce. The analogical scope here is structural: LLM agents do not accrue in-weights learning from a single conversation, but PROD_FAIL trajectories represent the kind of engagement that, in a human participant, would activate productive failure dynamics. Whether this generates equivalent transfer benefits in hybrid human-LLM settings is an open empirical question we leave for future work.

Limitations. Several limitations bound the current conclusions. First, all agents use the same underlying model (GPT-4.1); whether the framing effect generalises to other architectures, fine-tuned models, or models with different instruction-following proclivities is unknown. Second, our study is fully LLM-to-LLM; the presence of human interlocutors may substantially alter how framing instructions are interpreted and enacted, particularly if the human resists the learner-role dynamic. Third, the CPP rubric, while grounded in PISA, was adapted by our team and may not perfectly capture all dimensions of collaborative depth; future work should validate it against human-coded gold standards in larger samples. Fourth, the problem corpus is restricted to the six subjects of the MATH benchmark, and the generalisability of the CIDI split to other mathematical domains (combinatorics, proof-based mathematics) or to non-mathematical reasoning tasks remains to be established. Fifth, we cannot rule out that LLMs under C7 are merely performing collaborative language without the underlying reasoning structure that CPP assumes; micro-level trace analysis would be needed to address this.

7 Conclusion

We have shown that epistemic role framing—specifically, instructing LLM agents to adopt the identity of active learners—produces large, robust improvements in collaborative depth across all dimensions of the CPP framework. The effect size ($d = 1.30$ for CDI) is substantially larger than that produced by partner-awareness alone ($d = 0.16$, not significant after Bonferroni correction), and the advantage holds numerically across all 30 subject-difficulty cells of our problem corpus.

These findings carry two principal implications. For multi-agent system design, they suggest that system-prompt framing at the identity level should be treated as a first-class design variable, not an afterthought. For educational AI, they suggest that LLM agents can be made to simulate the kind of collaborative inquiry that human peer-learning research finds most beneficial—but that achieving this requires deliberate and specific epistemic role instructions.

Future work will extend this investigation in three directions: (1) increasing the number of agents beyond the dyadic setting to examine multi-way collaborative dynamics; (2) introducing human students as one participant in the pair to test whether the framing effects persist in human-AI collaboration; and (3) developing automated scoring of CPP metrics using LLM judges to scale annotation beyond human capacity.

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A CPP Annotation Rubric

Each of the 12 cells (P, C) for $P \in \{A, B, C, D\}$ and $C \in \{1, 2, 3\}$ is scored on a 0–3 scale. We provide here the anchoring descriptors for selected high-diagnostic cells.

Cell A1 (Exploring & Understanding / Shared Understanding). **0**: No evidence that agents jointly discussed the problem; one agent simply began solving. **1**: Agents acknowledged the problem exists and restated it, but no reconciliation of perspectives. **2**: Agents explicitly compared what each knows and flagged gaps; at least one question about the other's packet was asked and answered. **3**: Full joint problem model constructed; both agents articulate a shared interpretation that differs from what either could have produced alone, and this model visibly shapes the subsequent solution strategy.

Cell D3 (Monitoring & Reflecting / Team Organisation). **0**: No meta-discussion of the collaborative process or strategy; agents proceed to output without review. **1**: One agent flags a potential error but the flag is ignored by the partner. **2**: Agents engage in a substantive exchange about whether their approach is working; at least one revision of strategy is proposed. **3**: Agents explicitly evaluate the quality of their collaboration itself (e.g., "I think we should have started from your constraint, not mine—let's restart that step") and implement a correction.

Full rubric. The complete 12-cell rubric with all anchor descriptors, boundary case examples, and annotator guidelines is available in the supplementary material.

B CIDI Generation Prompt

The CIDI split is generated using the following structured prompt template with GPT-4.1 at temperature 0. The prompt instructs the generator to:

1. Identify the core mathematical objects and relations in the problem.
2. Partition those objects into two groups such that group A provides the structural constraints and group B provides the quantitative anchors (or a domain-appropriate analogue).
3. Verify that a solver with only group A information cannot determine the numerical answer even with unlimited mathematical knowledge, and vice versa.
4. Express each packet in natural language, avoiding explicit statements of what is absent from the other packet.

The generated split is validated by running a single GPT-4.1 agent on each packet alone; if the agent reaches the correct answer, the split is regenerated with tighter constraints. Typical regeneration rate was 12% of problems.

C Pilot Study: Condition Selection

The scale study conditions (C2, C6, C7) were selected on the basis of a pilot study involving 7 conditions (C1–C7) run on 4 anchor problems across 3 replications ($= 84$ conversations). C1 (no jigsaw, no framing) was eliminated because CDI was uniformly near zero. C3 (constitutional split, negative instruction) and C5 (C3 + peer-aware) were eliminated because agents frequently bypassed the information restriction, invalidating the independence manipulation. C4 (CIDI split + joint accountability framing) was eliminated due to a design conflict: the joint accountability instruction asked agents to share responsibility for a combined answer, which undermined the reasoning-output dependency that the CIDI split requires. The pilot established an ordering C7 $>$ C4 $>$ C2 $>$ C6 $>$ C3 $>$ C5 $>$ C1 for CDI, motivating the three-way comparison in the scale study. The unexpected result that C6 fell below C2 in the pilot (in the full study, C6

scores marginally above C2, $d = 0.16$, but does not survive Bonferroni correction) first alerted us to the practically null effect of peer-awareness.

D Subject-Level Generalisability Results

Table 4 reports mean CDI for each of the 30 subject $\times$ difficulty cells under conditions C2 and C7. C7 numerically outperforms C2 in all 30 cells. Cell n values range from 1 to 8 (mean 4.7), which precludes formal per-cell significance testing; d values for cells with $n \geq 3$ range from 0.63 to 2.51. Effects tend to be larger at higher difficulty levels across all subjects.

Table 5: Cell-level CDI means and Cohen's d (C7 vs. C2) for all 30 subject $\times$ difficulty combinations. d is omitted for $n < 3$ (insufficient data). Subject labels follow MATH benchmark categories [Hendrycks et al., 2021].

Subject	**Level**	n	**CDI (C2)**	**CDI (C7)**	d
Algebra	1	4	0.342	0.667	2.09
Algebra	2	3	0.333	0.711	2.51
Algebra	3	2	0.333	0.517	—
Algebra	4	7	0.386	0.557	1.38
Algebra	5	6	0.378	0.647	2.06
Geometry	1	5	0.360	0.483	1.25
Geometry	2	4	0.325	0.568	1.60
Geometry	3	6	0.305	0.536	1.19
Geometry	4	6	0.268	0.644	1.53
Geometry	5	6	0.306	0.619	2.29
Number Theory	1	1	0.367	0.833	—
Number Theory	2	5	0.326	0.546	0.73
Number Theory	3	6	0.290	0.496	2.77
Number Theory	4	5	0.364	0.659	0.90
Number Theory	5	8	0.409	0.669	0.91
Prealgebra	1	3	0.306	0.428	1.81
Prealgebra	2	6	0.403	0.581	1.14
Prealgebra	3	7	0.399	0.676	2.18
Prealgebra	4	4	0.336	0.558	1.30
Prealgebra	5	4	0.383	0.606	0.91
Precalculus	1	2	0.462	0.517	—
Precalculus	2	3	0.342	0.606	1.86
Precalculus	3	6	0.325	0.508	0.78
Precalculus	4	7	0.431	0.717	1.73
Precalculus	5	5	0.424	0.737	1.14
Probability	1	1	0.567	0.817	—
Probability	2	4	0.439	0.587	0.63
Probability	3	2	0.556	0.867	—
Probability	4	5	0.271	0.609	1.30
Probability	5	7	0.375	0.685	1.38